Technological innovation is the primary driver of economic growth in developed economies

Alexander Kruger

605298

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Full name: **Alexander Kruger**

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Technological innovation is defined as the creation of new methods, products, or processes through pioneering ideas. Quality and efficiency in the production of output are increased by technological innovations (Zhang, 2024)**. Technological innovations allow more to be produced for less. This is evident in, for example, the advancements in the Internet of Things (IoT), where simple tasks can be automated without constant labour costs. Another important aspect of technological innovation lies in the creation and reshaping of markets and companies therein, which drives investment.**

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The creation and reshaping of markets, resulting from technological innovations, have caused widespread economic growth. Research shows that the digital economy, e-commerce and the technology sector expanded their economic participation due to technological innovation (Ridwannudin, Rokhlinasari and Haerisma, 2025)**. In a similar vein, research shows that countries that can develop or adopt advanced technologies faster can gain a stronger position in the world market** (Ridwannudin, Rokhlinasari and Haerisma, 2025)**. However, the same logic can be applied to businesses. The faster a business can accept technological innovation, the stronger the resulting position it will have in a market. Regarding this occurrence, all businesses will compete to gain a competitive edge over their competition. Increased competition in markets stimulates economic growth because higher-quality products are being produced, and services are provided. Fundamentally, markets cause trade to occur; these transactions are a form of investments that stimulates economic growth. Research shows that innovation increases short- and long-term foreign direct investment (FDI)** (Dempere et al., 2023)**. Increased FDI accelerates economic growth and development. Additionally, the creation of new markets also drives local investment as new services and goods are available for use by the public. For example, the advancements in IoT brought with it a market for IoT. Due to the new market, products such as Sonoff switches became available to the public for the automation of lights and irrigation systems. Furthermore, the argument could be made that new markets created by technological innovation also drive investment into the employment sector.**

Even though a more skilled workforce is needed for newly created or reshaped companies, resulting in increased development costs, the benefit of technological innovation outweighs the negatives. Research shows two impacts of technological innovation on the labour markets (Su et al., 2022)**. The first impact is the eventual increased demand for labour due to the creation of new jobs. Secondly, it involves the replacement of human resources with high-tech alternatives. This phenomenon is considered a matter of controversy due to the loss of work opportunities and erosion of employment. Moreover, automation and the replacement of human resources save costs and stimulate economic growth. However, it's important to note that technological innovations such as automation and by extension, artificial intelligence can have a detrimental effect on the labour market. Evidently, technological innovation is accompanied by transitional friction. However, technological innovation ultimately leads to increased economic growth and increased demand for labour. Some jobs may vanish, but due to the nature of technological innovation, higher-paying jobs will replace the lost jobs** (Ridwannudin, Rokhlinasari and Haerisma,

Commented [KM2]: Which research are you referring to in this case? There is a disconnection between you previous idea on innovation with economic participation. Try and make your ideas flow.

Commented [KM3]: See my comments above.

Commented [KM4]: You could improve readability by dividing your script into paragraphs and subsections. This paragraph is too long.

Commented [KM5]: This phrase is over repeated in your script. Whose study is this ?

2025)**. The resulting workforce is more adaptable. However, for policymakers it's important to invest in education and not only technological innovation** (Ridwannudin, Rokhlinasari and Haerisma, 2025)

It is evident that technological innovation provides the means for countries and businesses to gain a competitive advantage. Furthermore, the improvement of existing markets and the creation of new markets, through technological innovation, drive investment. Capital invested in markets is exactly what ends up funding the improvement of the workforce and an improved human capital. Even if human resources are replaced with high-tech alternatives, the resulting increase in productivity with the eventual increase of labour demand further drives economic growth. Technological innovation is the way forward; by denying it a risk of economic stagnation appears. For nations, innovation is not only an option to grow, but it represents a survival strategy. Consequently, technological innovation is the engine transforms transitional friction into fuel for economic growth and expansion.

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